D2 - Data and Assumption Register

MIN E 413 Term Project - Pit I HG stockpile reconstruction (platform version 2.0, run 20261008T173954)

This register is the controlled record of:

1. the data received;
2. the quality findings, as a catalogue of QA rules with counts;
3. every preprocessing and cleaning decision;
4. every engineering assumption, with its need, justification, consequence and treatment.

Parameter values live in `config/project.yaml`, and each value there is tagged with the ID used here (`A-xx`). The QA results are in `outputs/qaqc/`, in machine-readable form.

No raw record was deleted or edited. Every input file is copied once into the managed raw store (`data/raw_store/`, named by SHA-256, logged in `ingest_log.csv`). The analysis reads only from the store. Corrections are made only through `config/manual_interventions.csv`, and every row there needs an id, type, target, reason, author and date.

Information type key, used throughout the project (course outline, Task 7): **M** = measured, **I** = inferred from data, **A** = assumed, **G** = model-generated.

1. Dataset inventory

ID	File(s)	Content	Coverage	Type	Notes
DS-1	`Dump Data.csv`	275 FMS truck-dump records: Cycle ID, payload (t), Au (g/t), Ag (g/t), Cu (%), density `Dens` (t/m³), dump X/Y/Z, timestamp (minute resolution)	2026-03-15 17:43 to 2026-03-31 20:52; 44,650 t	M	No truck ID or source (pit/bench) field. Grades are per **source signature** (21 distinct Cu/Au/Ag/Dens combinations), not per-load assays. The basis of `Dens` (bank or loose) is not documented
DS-2	`Topo_260315.dxf` … `Topo_260327.dxf` (8 files)	Vulcan DXF, 3DFACE triangles, one layer per file	Nominal dates 03-15 to 03-27 (none for 03-17 or 03-23 to 03-26)	M	Each file is a **closed change solid**: top = this survey, bottom = the previous surface, over the area that changed (QA-S01)
DS-3	`Topo_250328.dxf`, `Topo_250329.dxf`	As DS-2	Re-dated to 2026-03-28 and 2026-03-29 (MI-001, MI-002)	M	File-name year inconsistent with the record (rule SURVEY_NAME_YEAR)

ID	File(s)	Content	Coverage	Type	Notes
DS-4	(not supplied)	Full stockpile surface, pre-stockpile ground, density tests, dozer logs, truck IDs, moisture, survey acquisition times	None	n/a	Absence drives assumptions A-05, A-06, A-12, A-13, A-14, A-22

Units and coordinate reference. Coordinates are in metres. Easting is about 511,300 and Northing about 7,122,300, consistent with a UTM-type grid in the southern hemisphere. Elevation is about 4,480 m. The DXF headers say otherwise: all ten files declare `$INSUNITS = 1` (inches) and `$MEASUREMENT = 0` (imperial). This is a default template value, not the unit of the data, for three reasons:

- the coordinates coincide with the FMS mine grid (dumps sit on and behind the surveyed crests);
- in inches, the pile would be 2.5 m wide and stand 114 m above sea level;
- the volume budget (Section 2.5) balances recorded tonnes against surveyed volume only in metres; in cubic inches the solids would hold 1/61,000 of the FMS volume.

The header is therefore ignored (rule DXF_UNITS_HEADER). FMS and surveys overlay without offset, so **no transformation** was applied (A-01).

File dates. All ten DXF headers carry the same last-saved date, `$TDUPDATE` = 2025-11-14 00:52 local (07:52 UTC), which is earlier than the nominal survey dates. The layer names all carry the year token `25` (e.g. `250315_STOCK_HGHS`), and two file names do too. Together these suggest that the absolute dates were shifted by one year when the data set was prepared. Only relative timing matters to the model, and the FMS and survey sequences agree once MI-001/MI-002 are applied, so header dates are not used (rules DXF_SAVED_BEFORE_SURVEY, LAYER_DATE_MISMATCH).

2. QA/QC findings

2.1 QA rule catalogue

Every check is a named rule at one of three levels: record (FMS), survey file, or dataset. Each rule has a severity and a treatment:

- **error**: the record or file cannot be used as recorded. It is excluded from modelling and kept in the register, or it is corrected by a logged intervention.
- **warning**: modelled, with the treatment stated.
- **info**: recorded for traceability only.

Record-level rules are defined once (`spbm/rules.py`) and used by both the QA report and the engine. The table is generated from `outputs/qaqc/qa_rule_catalogue.csv`.

QA rule catalogue (counts from the run; tonnes for record rules)

rule_id	level	severity	count	tonnes	items	treatment
MISSING_PAYLOAD	record	error	0	0		Excluded from modelling, kept in register
MISSING_TIME	record	error	0	0		Excluded from modelling, kept in register
GRADE_MISSING	record	error	0	0		Excluded from modelling, kept in register
DENS_MISSING	record	error	0	0		Excluded from modelling, kept in register
LOC_MISSING	record	error	0	0		Excluded from modelling, kept in register
DUP_IDENTICAL	record	warning	0	0		First copy modelled, repeats excluded
DUP_CONFLICT	record	error	0	0		First copy modelled, others excluded; resolve with an fms_exclude intervention
PAYLOAD_NONPOSITIVE	record	error	0	0		Excluded from modelling
PAYLOAD_PARTIAL	record	info	3	152	cycle 666887, 667076, 671660	Modelled; payload error in Monte Carlo
PAYLOAD_HIGH	record	info	3	874	cycle 665064, 666376, 668024	Modelled; bounds the payload factor (A-06)
PAYLOAD_OFF_QUANTUM	record	warning	0	0		Modelled; checked manually
GRADE_INVALID	record	error	0	0		Excluded from modelling
GRADE_OUTLIER	record	info	1	114		Modelled (source-block grade); flagged
DENS_INVALID	record	error	0	0		Excluded from modelling

rule_id	level	severity	count	tonnes	items	treatment
LOC_OUTSIDE_DOMAIN	record	error	0	0		Location treated as unknown (as LOC_DEFAULT)
LOC_DEFAULT	record	warning	14	2,356	cycle 664887, 664959, 665073, 666320, 666394, 666376, 666430, 666471, 666564, 666692, 6…	Location treated as unknown: uniform push cost over the interval's fill
LOC_OUTSIDE_SURVEYS	record	info	56	9,120		Modelled: pushed from the dump point (crest dumping)
LOC_Z_INCONSISTENT	record	info	2	380	cycle 667102, 671466	Modelled; dump Z is not used by the engine (X/Y only)
TIME_BEFORE_FIRST_INTERVAL	record	error	0	0		Excluded (belongs to the pre-record pile)
TIME_AFTER_LAST_SURVEY	record	warning	1	228	cycle 673291	Placed provisionally by the forward model; reconciled when the next survey arrives
TIME_DUPLICATE	record	info	2	304	cycle 666880, 666882	Modelled; both kept
ID_OUT_OF_ORDER	record	info	33	5,358		None - the timestamp governs ordering
IN_SURVEY_GAP_INTERVAL	record	info	63	9,918		Modelled; the interval's spatial resolution in time is coarser

rule_id	level	severity	count	tonnes	items	treatment
DXF_UNITS_HEADER	survey	warning	10		all files	Header ignored: coordinates are metres (they match the FMS mine grid; in inches the pile would be 0.1 km above sea level)
DXF_SAVED_BEFORE_SURVEY	survey	info	10		all files (TDUPDATE 2025-11-14)	Header dates not used for timing; files were exported in one batch
SURVEY_NAME_YEAR	survey	error	2		Topo_250328.dxf, Topo_250329.dxf	Re-dated by manual intervention (MI-001, MI-002)
LAYER_DATE_MISMATCH	survey	info	8		Topo_260315.dxf, Topo_260316.dxf, Topo_260318.dxf, Topo_260319.dxf, Topo_260320.dxf, To…	File name (with interventions) governs; month/day agree
ZERO_PLAN_AREA_FACES	survey	info	10		all files (214-639 per file)	Ignored by the rasteriser (they carry no top/bottom elevation)
DEGENERATE_FACES	survey	info	8		Topo_260315.dxf, Topo_260318.dxf, Topo_260319.dxf, Topo_260320.dxf, Topo_260322.dxf, To…	Ignored
DUPLICATE_FACES	survey	info	6		Topo_260315.dxf, Topo_260318.dxf, Topo_260319.dxf, Topo_260320.dxf, Topo_260322.dxf, To…	No effect (per-cell max/min)
SOLID_NOT_CLOSED	survey	warning	0			Single-surface cells take thickness 0; checked against previous surface

rule_id	level	severity	count	tonnes	items	treatment
NORMALS_INVERTED	survey	info	1		Topo_250328.dxf	No effect (orientation-free rasteriser)
SOLID_BASE_MISMATCH	survey	warning	1		Topo_260318.dxf	Solid bottom trusted (A-13b off); A-13b sensitivity
SURVEY_GAP	dataset	warning	2		interval 2 (03-17 07:00 to 03-19 07:00, 48 h); interval 7 (03-23 07:00 to 03-28 07:00, ...	Coarser time resolution; listed in reconciliation
FMS_PARTIAL_FIRST_INTERVAL	dataset	warning	1		first FMS record 2026-03-15 17:43, interval 0 starts 2026-03-15 07:00	Interval 0 excluded from timing and volume calibration; unexplained fill is unattributed
FMS_SILENCE	dataset	warning	2		interval 2 (38 h); interval 7 (64 h)	Interval checked in the volume calibration
VOLUME_RATIO_OUTLIER	dataset	warning	3		interval 0 (ratio 2.35); interval 7 (ratio 2.32); interval 9 (ratio 2.17)	Excluded from the f_v calibration; the unexplained fill is unattributed volume
SURVEY_CHANGE_NO_FMS	dataset	warning	0			Fill is unattributed
PAYLOAD_QUANTISED	dataset	info	1		275 records	Payload treated as an estimate (A-10): per-load CV and systematic payload factor
GRADE_UNITS_UNCONFIRMED	dataset	warning	1		mean Cu 8.42 %, Au 8.00 g/t	Units used as supplied; flagged in the executive summary (QA-F04)

40 rules are checked and 25 are triggered by the supplied data. No FMS record meets an error rule, so all 275 records are modelled. The 14 LOC_DEFAULT records (2,356 t) are modelled with an unknown location.

2.2 FMS record findings (DS-1)

ID	Finding	Evidence (rule)	Action / treatment	Influence on model
QA-F01	No missing, invalid or duplicated fields; no repeated Cycle IDs	Error rules = 0	None	-
QA-F02	**Payload is not weighed.** All 275 payloads are exact multiples of 38 t (38-304 t; 152 t = 155 loads), so payload = passes × nominal bucket	PAYLOAD_QUANTISED, PAYLOAD_OFF_QUANTUM = 0	Per-load CV 8% (A-10); a *systematic* payload factor f_pay 1.00-1.19 (A-06). The largest load (304 t) sets the upper bound	Tonnage and metal uncertainty
QA-F03	**Grades are source-block values**, not load assays: 21 signatures. Cu 5.6-13.7%, Au 5.6-10.1 g/t, Ag 147-256 g/t, Dens 1.92-2.22 t/m³	`grade_signatures.csv`	Best available grade per load. Grade error per signature (correlated across loads of a source), CV 10% (A-11)	Within-load homogeneity assumed
QA-F04	Grade magnitudes are very high for a Cu-Au mine (mean 8.4% Cu, 8.0 g/t Au)	GRADE_UNITS_UNCONFIRMED	Units used **as supplied**, labelled in every output, stated in the executive summary	Absolute metal values depend on unit confirmation; relative patterns do not
QA-F05	33 Cycle IDs out of chronological order; 2 records share a timestamp (03-19 21:40, different positions)	ID_OUT_OF_ORDER, TIME_DUPLICATE	Information only. The timestamp defines order	None
QA-F06	**Dump Z is unreliable.** Where the cell had been surveyed before the dump, Z minus that surface has median −0.3 m and P10/P90 −2.5/+3.6 m (n = 197). Two dumps lie outside the [interval start, interval end] surface envelope by more than 5 m	LOC_Z_INCONSISTENT = 2	Z not used by the algorithm; only X/Y is used (A-29)	Removes a noisy input

ID	Finding	Evidence (rule)	Action / treatment	Influence on model
QA-F07	**FMS coverage by production day** (D 07:00 to D+1 07:00, `daily_coverage.csv`). 03-17 has neither FMS records nor a survey. Its fill appears in the 03-18 solid, and that interval's volume ratio (1.40) is consistent, so **03-17 was a non-dumping day, not an FMS outage**. Genuine FMS gaps (dataset rules FMS_SILENCE, VOLUME_RATIO_OUTLIER): • interval 0, where the record starts at 17:43 on 03-15; • interval 7 (03-23 07:00 to 03-28 07:00), with 21 loads, silences of 65 h and 64 h, and a ratio of 2.32; • interval 9, where the record stops at 08:17 on 03-29 (ratio 2.17)	`volume_factor_calibration.csv`	These three intervals are excluded from the volume calibration. Their fill not explained by FMS becomes **unattributed** (A-21)	2,869 m³ unattributed
QA-F08	1 load (228 t, 03-31 20:52) is after the last survey cut-off (03-30 07:00)	TIME_AFTER_LAST_SURVEY	Placed **provisionally** with the forward model; re-reconciled when the next survey arrives	0.5% of tonnes
QA-F09	56 dumps (9,120 t) lie outside the union of all change solids (median 8 m, max 26 m): dumping on the unsurveyed platform behind the crest	LOC_OUTSIDE_SURVEYS	Modelled; pushed from the dump point. None is more than 100 m away (LOC_OUTSIDE_DOMAIN = 0)	Push distances
QA-F10	**Default / stuck FMS position.** 14 records (2,356 t, 5.3% of FMS tonnes, 03-16 to 03-28) share the identical position E 511,334.7, N 7,122,320.4, Z 4,473.0. That point is below the floor and, until 03-27, in front of the toe	LOC_DEFAULT	Location **unknown**: the load keeps its tonnes, grade and interval, but no position. Its push cost is uniform over the interval's fill (A-32). Version 1 imputed the period centroid instead, which is still an invented position	Allocation of 2.4 kt within its intervals

2.3 Surveys (DS-2, DS-3)

ID	Finding	Evidence	Action / treatment	Influence
QA-S01	**Each DXF is a closed change solid, not a surface.** 100% of cells are crossed by exactly two faces. The divergence-theorem volume equals the gridded volume (e.g. 2,868 vs 2,867 m³). The bottom of solid *s* equals the top of solid *s−1* wherever they overlap (P5-P95 = 0.000 m)	`survey_qaqc.csv` (SOLID_NOT_CLOSED = 0)	Top = surface at survey *s*; bottom = surface before it; only changed areas are present	Gives measured fill per interval (M)
QA-S02	File names `Topo_250328`/`Topo_250329` carry year 2025. The FMS shows dumping on 2026-03-28/29, the faces advance beyond the 2026-03-27 solid, and their bottoms match its top	SURVEY_NAME_YEAR	Re-dated by **MI-001, MI-002** (reason, author, date recorded; files not renamed)	Chronology
QA-S03	DXF header: `$INSUNITS = 1` (inches), `$MEASUREMENT = 0` (imperial), `$TDUPDATE` 2025-11-14 in all files	DXF_UNITS_HEADER, DXF_SAVED_BEFORE_SURVEY	Header ignored; data are metres (Section 1)	None, once recognised
QA-S04	**Survey acquisition time not supplied**	`temporal_alignment.csv`, `dump_time_of_day.csv`	Cut-off D+1 07:00 (A-05, Section 2.4)	Assignment of loads to intervals
QA-S05	Missing surveys: no file for 03-17 or 03-23 to 03-26. The bottom of the 03-18 solid differs from the previous top on 23% of overlapping cells	SURVEY_GAP, SOLID_BASE_MISMATCH	Intervals flagged; the solid bottom is trusted (A-13b off)	Coarser time resolution in intervals 2 and 7
QA-S06	**Elevation decreases are not stored as solids.** In the 03-18 solid, 22% of overlap cells have bottoms up to 1-2 m *below* the earlier surveyed surface (dozer cuts, slumping); 6% in the 03-28 solid	`period_reconciliation.csv` (`cut_volume`)	Treated as cuts: the column top is removed and re-handled as a transport source (A-20). 565 m³ in total	Mixing of earlier material into later fill
QA-S07	4 surveys (03-16, 03-21, 03-22, 03-29) do not reach the floor (minimum Z 4,476.4-4,478.8 m vs floor about 4,475 m): the change polygon was clipped above the toe	`reaches_floor`	Recorded; the volume budget shows no missing fill on complete days (A-13b off)	Sensitivity S2
QA-S08	`Topo_250328.dxf` has inverted face winding (signed volume −4,924 m³). All files contain vertical faces with zero plan area (214-639 per file). Most contain a few degenerate (≤ 25) and duplicate (≤ 6) faces	NORMALS_INVERTED, ZERO_PLAN_AREA_FACES, DEGENERATE_FACES, DUPLICATE_FACES	The rasteriser is winding-independent (top = max, bottom = min); faces without plan area are skipped	None

ID	Finding	Evidence	Action / treatment	Influence
QA-S09	Triangle edge median 0.93-1.15 m, so the native resolution is about 1 m	`edge_median_m`	Simulation grid 1 m (A-24)	Resolution
QA-S10	Face slopes (cells > 25°): median 36.7°, mode 37.0°, P75 38.5°	`fig_qaqc_repose.png`	Angle of repose 37° (A-03)	Forward model, inference
QA-S11	Toe elevations define a near-horizontal floor: plane 4,475.08 m, dip 0.29°, residual sd 0.26 m	`fig_qaqc_floor_plane.png`	Floor plane (A-12)	Pre-record core thickness
QA-S12	Survey precision cannot be estimated from repeat surveys (bottoms are copies of previous tops). 8-16% of solid area is thinner than 0.1 m	`sliver_frac_lt_0.1m`	σz = 0.05 m assumed; detection limit τ = 0.14 m; changes below τ are not moved	Negligible volume

2.4 Survey timing (A-05)

The survey cut-off was scanned from 0 to 36 h after 00:00 of the nominal survey date, in 0.5 h steps. For each cut-off, the loads were re-assigned and the interval FMS volume Σ T/Dens was compared with the surveyed solid volume. The comparison uses intervals between consecutive-day surveys, a set that does not depend on the cut-off.

Survey cut-off scan (selected rows of outputs/growth/temporal_alignment.csv)

cut-off (h after 00:00 of D)	clock time	r (consecutive-day intervals)	robust CV of ratio	r (all intervals after the first)	loads after last survey
0	D+0 00:00	-0.469		-0.461	19
6	D+0 06:00	0.039	0.73	0.013	8
12	D+0 12:00	0.396	0.88	0.253	1
18	D+0 18:00	0.610	0.42	0.372	1
21	D+0 21:00	0.706	0.33	0.422	1
24	D+1 00:00	0.961	0.23	0.720	1
27	D+1 03:00	0.950	0.23	0.768	1
29	D+1 05:00	0.974	0.23	0.927	1
30	D+1 06:00	0.992	0.08	0.977	1
31	D+1 07:00	0.992	0.08	0.976	1
31.5	D+1 07:30	0.992	0.09	0.976	1
32	D+1 08:00	0.981	0.10	0.965	1
34	D+1 10:00	0.971		0.951	1
36	D+1 12:00	0.959		0.912	1

The scan has a plateau from 30 to 31.5 h (r = 0.992, robust CV 0.08). Midnight (24 h) gives r = 0.961 and robust CV 0.23. The dump times corroborate the result. Only 2 of 275 dumps fall between 06:00 and 07:30, and none falls between 18:30 and 20:00 or between 00:00 and 01:30. These are shift changes and meal breaks. Any cut-off between 06:00 and 07:30 therefore gives the same assignment. **The cut-off is D+1 07:00 (31 h)**, the production-day convention. Version 1 searched only 21:00-23:59 of day D, and its "optimum" of 23:59 was the end point of that range.

2.5 FMS vs survey volume reconciliation and the volume factor (A-06)

The surveyed change solids total **35,096 m³**: fill 35,108 m³, less 565 m³ of revealed cuts, gives a net fill of 34,543 m³. Recorded FMS tonnage converts to 21,525 m³ at the FMS densities (Σ T/Dens). Interval by interval:

Volume budget per survey interval (interval of survey D ends at D+1 07:00)

interval	survey (D)	fill m³	cut m³	net fill m³	FMS t	Σ T/Dens m³	longest FMS silence (h)	ratio	used for f_v	role in hold-out	hold-out error
0	03-15	2,853	0	2,853	2,432	1,212	11	2.35	no (FMS gap)	calibration	
1	03-16	2,300	0	2,300	4,142	1,931	19	1.19	yes	calibration	
2	03-18	4,792	317	4,475	6,802	3,188	38	1.40	yes	calibration	
3	03-19	8,699	0	8,699	11,704	5,495	6	1.58	yes	calibration	
4	03-20	2,143	0	2,143	3,040	1,427	11	1.50	yes	calibration	
5	03-21	1,420	4	1,416	2,166	1,019	13	1.39	yes	held-out	-5.3%
6	03-22	2,908	57	2,851	3,838	1,939	13	1.47	yes	held-out	+0.5%
7	03-27	3,770	13	3,757	3,116	1,617	64	2.32	no (FMS gap)	held-out	+37.0%
8	03-28	4,928	173	4,755	6,004	3,101	11	1.53	yes	held-out	+4.6%
9	03-29	1,295	0	1,295	1,178	596	23	2.17	no (FMS gap)	held-out	+32.7%

On the seven intervals with complete FMS records (consistent: within median ± 3 robust SD), measured net fill is a stable 1.19-1.58 times Σ T/Dens. The pooled value is the **volume factor f_v = 1.472** (bootstrap P10-P90 1.40-1.52). The three excluded intervals are the FMS gaps of QA-F07, at ratios 2.2-2.4.

f_v is predictive, not merely fitted:

Volume factor hold-out tests

Hold-out test (`volume_factor_summary.csv`)	Mean absolute volume error on held-out consistent intervals
f_v calibrated on the first half (intervals 0-4), predicting intervals 5, 6 and 8	**3.5%**
Constant density (tonnes ÷ volume) calibrated on the first half	8.7%
Dens used as the in-pile density (f_v = 1)	32%

Hold-out test (`volume_factor_summary.csv`)	Mean absolute volume error on held-out consistent intervals
Leave one consistent interval out	8.0%

What f_v means is **not identifiable** from these data. The in-pile volume of a load is T × f_v / Dens. Two explanations, and any mix of them, fit the same volume:

- **bulking**: the in-pile density is below Dens. If Dens is a bank density, f_v = 1.47 is an ordinary swell factor for blasted rock;
- **under-recorded or unrecorded tonnes**: either payloads understated (pass-count × nominal bucket) or whole loads missing.

The platform carries the split as the payload factor f_pay (true tonnes = T × f_pay, in-pile density = Dens × f_pay / f_v). The base case is f_pay = 1 (recorded tonnes; in-pile density 1.40 t/m³). The Monte Carlo samples f_pay from 1.00 to 1.19 (triangular, most likely 1.00), where the upper bound takes the largest load of 304 t to 362 t, about the largest haul-truck rating. Two interpretations fall outside that range:

- **X1** (all of the excess is payload under-recording). This is excluded, because the largest load would be 448 t.
- **X2** (the excess is loads missing from the FMS, Dens being the in-pile density). This cannot be excluded from these data. It is reported as a separate scenario, “not included in the ranges” (D6, Section 10).

If Dens is a loose (truck-box) density, bulking by a further 47% is implausible, and X2 or under-recording becomes the more likely explanation. **Confirming the basis of Dens is the single most valuable data item for the tonnage estimate** (Section 5).

3. Preprocessing decisions

ID	Decision	Why necessary	Action	Possible influence	How original is retained
PP-01	Survey re-dating	Chronology	Interventions MI-001, MI-002 (reason, author, date)	Wrong order would corrupt sequencing	File names unchanged; `date_from_name` and `assigned_date` both stored; intervention id in the survey register
PP-02	Survey cut-off D+1 07:00	Assign loads to intervals	Config value (`time.survey_cutoff_hours: 31`), chosen by the cut-off scan	Loads near the cut-off shift one interval	`temporal_alignment.csv`, `dump_time_of_day.csv`
PP-03	Rasterise solids to a 1 m grid (top/bottom at cell centres)	Common computational grid	Cached by file hash	≤ 1 m edge effects	Raw DXF untouched
PP-04	Dump Z ignored	Unreliable (QA-F06)	Only X/Y used	None	Z kept in the loads table

ID	Decision	Why necessary	Action	Possible influence	How original is retained
PP-05	Unattributed fill volume	FMS gaps (QA-F07)	Separate material class, imputed grade	2,869 m³ with imputed grade	Class tonnage per block
PP-06	Cuts detected from bottoms below earlier tops	Solids omit decreases	Removal + re-handling	Mixing	`cut_volume` per interval
PP-07	Load after the last survey	Not yet surveyed	Forward-model nowcast, flagged	228 t	Class “Provisional”
PP-08	No record excluded	Brief: do not simply delete	Error rules exclude from modelling only; none triggered	-	All 275 records modelled and kept in `fms_records_with_flags.csv`
PP-09	Default-position records modelled with unknown location	Position physically impossible (QA-F10)	Uniform push cost over the interval’s fill	Their material is spread where located loads do not claim the fill	`x_raw` / `y_raw` retained; `loc_unknown` in `movement/loads_used.csv`
PP-10	DXF header units ignored	Header says inches; data are metres (QA-S03)	Coordinates read as metres	None	Header values in `survey_qaqc.csv`

4. Assumption register

Columns follow the course outline, Task 4: Assumption | Need | Justification | Consequence if wrong | Treatment. **S** = tested in sensitivity (one-at-a-time); **MC** = sampled in the Monte Carlo; **Sc** = reported as a scenario.

ID	Assumption (value / rule)	Need	Justification	Consequence if wrong	Treatment
A-01	Coordinates of FMS and surveys share one projected mine grid (m); no transformation; DXF header units ignored	Overlay data	Dumps lie on and behind surveyed crests; consistent elevations; header unit implausible (QA-S03)	Systematic offset would bias push distances	Visual + statistical check (QA-F06, F09)
A-02	`Topo_2503xx` files are 2026 surveys	Chronology	QA-S02 evidence (FMS dates, face progression, bottom = previous top)	Sequence error	Interventions MI-001/002

ID	Assumption (value / rule)	Need	Justification	Consequence if wrong	Treatment
A-03	Angle of repose 37°	Forward model; inference of unseen faces	Measured mode 37.0°, median 36.7° of surveyed faces; typical 35-40° for blasted rock	Shape of inferred faces and nowcast	S (forward calibration 35/37/39°)
A-04	DXF solids record accretion only; bottom = previous surface	Derive fill per interval	QA-S01, QA-S06	Misread change	Verified on all 9 overlaps
A-05	A survey labelled D closes its interval at D+1 07:00 (31 h after 00:00 of D); the first interval is 24 h long	Assign loads to intervals	Cut-off scan plateau 30-31.5 h; shift-change lull 06:00-07:30 (Section 2.4)	Loads shift one interval	S (29/32 h), MC (29-32 h, each with its own f_v calibration)
A-06	In-pile volume = T × f_v / Dens, f_v calibrated (auto, 1.472); true tonnes = T × f_pay, base f_pay = 1.0 (excess volume = bulking; in-pile density 1.40 t/m³)	Convert tonnes to volume and back	f_v hold-out error 3.5% vs 32% for f_v = 1 (Section 2.5). The bulking/under-recording split is not identifiable	Record-period tonnes +47% if the excess is missing loads (X2)	MC (f_v bootstrap; f_pay triangular 1.00-1.19, mode 1.00); S (f_v 1.35/1.60, f_pay 1.19); Sc (X1 excluded, X2 not included)
A-07	Truck GPS horizontal error σ = 3 m	Uncertainty	Typical HP GPS on haul trucks 1-5 m; dump-event latency	Destination of individual loads	MC
A-08	Survey vertical precision σz = 0.05 m	Detection limit, uncertainty	Typical drone photogrammetry/LiDAR on rock piles 0.03-0.10 m	Noise counted as fill	S (0.025/0.15), MC (per-survey bias 0.05 m)
A-09	Detection limit τ = 2·√2·σz = 0.14 m	Ignore noise	Standard LoD for DEM differencing (95%)	Small volume not moved	Robustness (k = 1 to 4)
A-10	Payload error CV 8% per load; payload = passes × 38 t	Tonnage uncertainty	Pass-count payload (QA-F02); industry payload-monitor accuracy ±5-10%	Tonnes and metal	MC (systematic part via f_pay, A-06)
A-11	Grade of a load = its source signature; homogeneous within a load; signature error CV 10%	Assign grade	Only data available (QA-F03)	Block grade variance under-represented	MC (correlated by signature)
A-12	Floor = plane fitted to toe cells (4,475.08 m, 0.29° dip)	Bottom of model	Toe elevations consistent across surveys (sd 0.26 m)	Pre-record core volume ±0.26 m × area	Reported; floor residual map
A-13	Never-surveyed cells: behind the crest = flat platform at the nearest crest elevation; in front of the face = face continued at 37° to the floor	Geometry where no survey exists	Truck-operated platform is flat; faces at repose	Pre-record core volume and extent	Geometry class “inferred”; reported separately

ID	Assumption (value / rule)	Need	Justification	Consequence if wrong	Treatment
A-13b	**Off in the base case.** (If on: first-surveyed cells in front of an earlier face were at its repose extrapolation; a solid bottom above it is fill not captured by any solid.)	Option for toe-clipped solids	With f_v calibrated, recorded loads explain measured fill of complete days to ~3%, leaving no room for ~7,200 m³ of unrecorded fill	About 7,200 m³ moves from the pre-record core to unattributed	S (switch); Sc (S2)
A-14	Model domain = convex hull of surveyed change and dump locations	Lateral limit of the pile	No full-pile survey supplied	The back of the pile outside the hull is not modelled	Stated limitation; R-01
A-15	Within an interval's measured fill, dozer push follows least total push work (optimal transport over a slope-aware geodesic cost)	Unknown dozer trajectories	Operators minimise push distance; OT is the classical earth-moving formulation. **Evidence is weak**: in the twins the push cost helps only when the truth follows the same physics, and a flat cost (no push term) does better otherwise	Block grades change by about 0.2% Cu on average (flat vs geodesic); totals unchanged	MC samples geodesic/flat 50/50; tornado; twin ablation; tracer trials (R-06)
A-16	Uphill push penalty α = 4 m per m of lift	Prevent implausible uphill pushes	Dozer production falls sharply uphill (grade correction factors, Caterpillar Performance Handbook). The twin with α = 0 is no worse	Allocation across the crest	S (1/10), MC U(1, 10), twin (α = 0)
A-17	Maximum effective push distance 90 m (geodesic)	Limit movement	Dozing is generally economical to about 100 m	Loads may be retained on the platform	S (50/130), MC U(50, 130)
A-18	Mixing length ε = 6 m (entropy regularisation)	Degree of mixing	GPS error (3 m), blade spill, rolling on the face; twin favours small-moderate ε	Over- or under-mixing of grades	S (2/15), MC U(2, 15), twin
A-19	Within one interval, the fill of a column is vertically homogeneous	Vertical placement	Daily fill is built by many interleaved loads	Vertical grade resolution limited to the interval layer (0.5-3 m)	Inherent; documented
A-20	Elevation decrease below an earlier surface = material removed by dozer/slumping and re-handled in the same interval	Solids omit cuts (QA-S06)	Dozers trim crests while pushing	Cut volume re-mixed	Reported per interval

ID	Assumption (value / rule)	Need	Justification	Consequence if wrong	Treatment
A-21	Unattributed fill has the tonnage-weighted mean grade and in-pile density of FMS loads in the same and adjacent intervals	FMS gaps (QA-F07)	Material placed in the same days most likely came from the same faces	Grade of 3,901 t is imputed	MC bootstrap of grades; separate class
A-22	Pre-record core (legacy): in-pile density and grade = FMS mean	Populate blocks below the pre-record surface	No information supplied	89,795 t carry an **assumed** grade	Class “Legacy-assumed”, confidence “Uninformed”; reported after the record-period inventory, never as measured
A-23	FMS volume not seen in an interval’s fill is retained on the platform and pushed later (dozer lag)	Timing mismatch	Paddock dumping and pushing later is normal practice	Grade appears in the next interval’s fill	Carried volume reported
A-24	Simulation grid 1 m, vertical integration exact (layers)	Numerics	Matches survey triangle size (QA-S09)	Edge effects	Robustness 0.5/2.5 m
A-25	Block size 5 × 5 × 2 m, partial blocks kept with fill fraction	Planning unit	Information resolution about 5 m (GPS + mixing); 2 m gives several layers per face height and suits loader slices	Smoothing of grade	S (2 × 2 × 1, 10 × 10 × 4)
A-26	Tonnes as reported by the FMS; no moisture correction (wet/dry basis unknown)	Units	No moisture data	Systematic tonnage bias of a few %	Stated; R-05
A-27	No particle-size segregation of grade down the face	Not observable	No size-grade data	Toe vs crest grade bias	Stated limitation
A-28	No time-dependent settlement after placement	Not observable	Short record (2 weeks)	Small volume bias	Absorbed in the f_v calibration
A-29	Dump Z is not used	QA-F06	Unreliable	None	-
A-30	Boundary conditions: material cannot leave the domain; outside the domain is not modelled	Numerics	Domain covers all observed change	None inside the domain	-
A-31	Missing data: FMS gaps → unattributed class; survey gaps → gap intervals; no synthetic loads are invented	Missing-data treatment	Transparency over completeness	Unattributed share	Reported; R-02

ID	Assumption (value / rule)	Need	Justification	Consequence if wrong	Treatment
A-32	Records sharing an identical X/Y/Z with ≥ 2 others (LOC_DEFAULT), or > 100 m from every solid (LOC_OUTSIDE_DOMAIN), have an **unknown** dump location: uniform push cost over their interval's fill	14 records have an impossible location (QA-F10)	Identical coordinates to 0.01 m over 12 days, below the floor	Allocation of 2.4 kt within its intervals	Unit and real-data tests; `fms_location_unknown` intervention for further cases

5. Data that would most reduce uncertainty (value of information)

1. **The basis of `Dens`** (bank or loose density) and a bulk-density test on the pile. This decides between bulking and missing tonnes (A-06), and with it record-period tonnes within a factor of 1.472.
2. A full-pile survey (current) and a pre-stockpile ground survey. These remove A-13 and A-14 and most of A-22.
3. An FMS completeness audit for 03-23 to 03-27 and after 03-29 08:17 (removes most unattributed volume), and the payload calibration of the loader buckets (f_pay).
4. Survey acquisition times (A-05).
5. Truck ID, source bench and per-load grade (A-11).
6. Tracer trials (RFID or coloured tags in selected loads) to calibrate ε and the push model (A-15 to A-18).